Output-error sensitivity of a Clements mesh vs. effective insertion-loss spread (small range; fixed $N = 100$, varying μ_L)

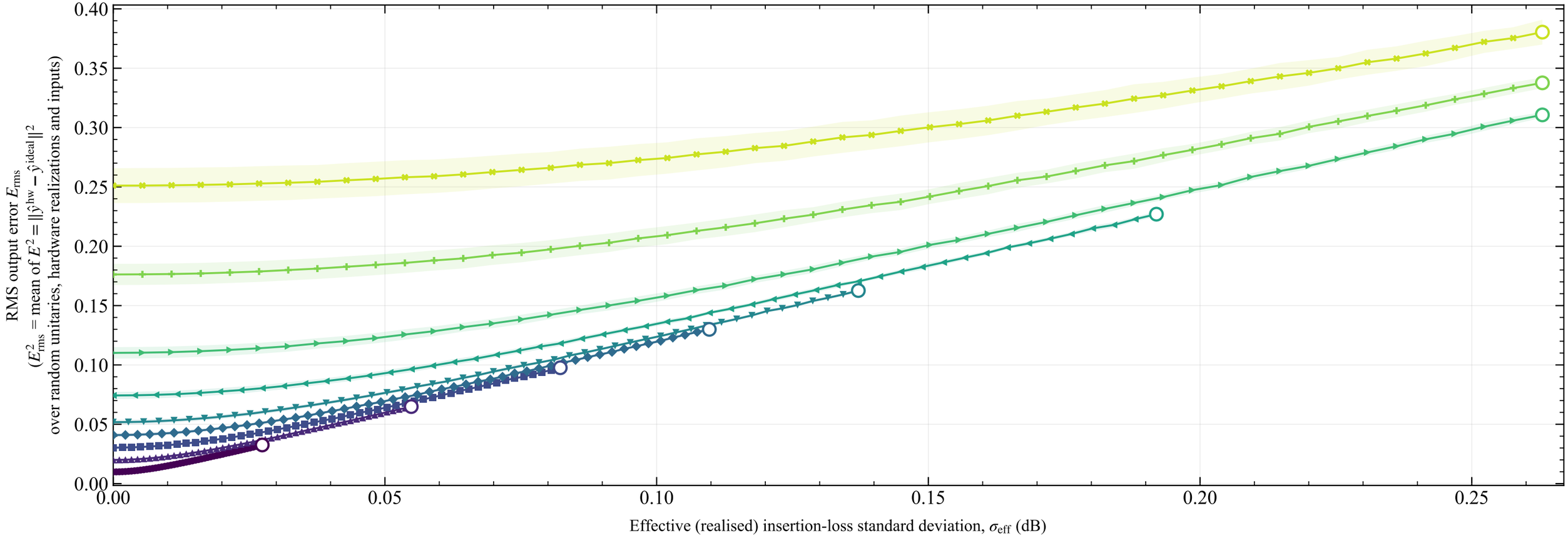
$N = 100$ | Haar-random unitary U (single Clements mesh) | one forward pass $y = Ux$ | $\hat{y}$ = complex output field, normalised to unit L2 norm | $E = \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2$

Inputs per unitary: 7 types x 10 random draws (dense complex, dense real, binary ±1, phase-only, sparse 10%, single port, Gaussian beam) | shown: mean over all 7 types (equal weight)

30 hardware realizations per unitary per point (each chip sees one input of every type; each input used on 3 chips) | 25 Haar unitaries | 50 points per curve in $\sigma_{\mathrm{eff}} \in [0, \sigma_{\mathrm{top}}(\mu_L)]$ dB | fixed $N = 100$, one curve per mean loss μ_L

Hardware model: per-arm loss $L \sim \mathrm{TruncNorm}(\mu_L, \sigma_L^2; 0, 2\mu_L)$ dB, amplitude $10^{-L/20}$, independent for each arm of every MZI and chip; swept: realised std σ_{eff} of L (σ_L by inversion); single Clements mesh, ideal phase shifters, 50:50 couplers and output phase screen

Each point: $E_{\mathrm{rms}} = \sqrt{\langle E^2 \rangle}$ over unitaries, chips and input types; band: ±1 s.d. across Haar-random targets (s.e. in the CSV) | $\sigma_{\mathrm{eff}} = 0$: delta at μ_L (starts at $E_L(\mu_L, N)$) | $\sigma_{\mathrm{top}} = \min(0.95\,\mu_L/\sqrt{3}, 0.263)$ dB



Mean loss μ_L (fixed per curve) | $E_0 = E_{\mathrm{rms}}(\sigma_{\mathrm{eff}} = 0)$, $E_{\max} = E_{\mathrm{rms}}$ at σ_{top} (open circle = end of curve); $\sigma_{\mathrm{top}} = \min(0.95\,\mu_L/\sqrt{3}, 0.263$ dB)

- $\mu_L = 0.05$ dB ($\sigma_{\mathrm{top}} = 0.02742$ dB) ($E_0 = 0.010$, $E_{\max} = 0.033$)
- $\mu_L = 0.1$ dB ($\sigma_{\mathrm{top}} = 0.05485$ dB) ($E_0 = 0.020$, $E_{\max} = 0.065$)
- $\mu_L = 0.15$ dB ($\sigma_{\mathrm{top}} = 0.08227$ dB) ($E_0 = 0.030$, $E_{\max} = 0.098$)
- $\mu_L = 0.2$ dB ($\sigma_{\mathrm{top}} = 0.1097$ dB) ($E_0 = 0.041$, $E_{\max} = 0.130$)
- $\mu_L = 0.25$ dB ($\sigma_{\mathrm{top}} = 0.1371$ dB) ($E_0 = 0.052$, $E_{\max} = 0.163$)
- $\mu_L = 0.35$ dB ($\sigma_{\mathrm{top}} = 0.192$ dB) ($E_0 = 0.074$, $E_{\max} = 0.227$)
- $\mu_L = 0.5$ dB ($\sigma_{\mathrm{top}} = 0.263$ dB) ($E_0 = 0.110$, $E_{\max} = 0.311$)
- $\mu_L = 0.75$ dB ($\sigma_{\mathrm{top}} = 0.263$ dB) ($E_0 = 0.176$, $E_{\max} = 0.338$)
- $\mu_L = 1$ dB ($\sigma_{\mathrm{top}} = 0.263$ dB) ($E_0 = 0.251$, $E_{\max} = 0.380$)